



Labradorite

△ **Labradorite rough** in typical gemstone-grade base colour

A type of feldspar, labradorite is named for the Canadian province of Labrador, where it was first identified in 1770. Gemstone labradorite is commonly characterized by its rich play of iridescent colours, principally blue, on broken surfaces. Crystals that display this effect are cut *en cabochon* or used in carvings. Nearly transparent material with a beautiful iridescence comes from southern India. Fully transparent labradorite is found from time to time, and can be yellow, orange, red, or green.

Specification

Chemical name Sodium, calcium aluminosilicate | **Formula** $\text{NaAlSi}_3\text{O}_8 - \text{CaAl}_2\text{Si}_2\text{O}_8$ | **Colours** Blue, grey, white
Structure Triclinic | **Hardness** 6–6.5 | **SG** 2.65–2.75
RI 1.56–1.57 | **Lustre** Vitreous | **Streak** White | **Locations** Madagascar, Finland, Russia, Mexico, USA; Labrador, Canada



Labradorite in combination | **Rough** | This piece of labradorite rough shows gem-quality blue material interlayered with another feldspar.



Square cabochon | **Cut** | This labradorite cabochon has fine blue, gold, and green schiller. Material like this is found in Mexico and the USA.



Animal carving | **Carved** | The schiller in labradorite, if properly oriented, adds depth and life to carvings, as here. It combines with the vitreous lustre of the mineral to give a glowing, greenish surface sheen reminiscent of the slimy skin of a frog.



Cameo head | **Carved** | Skilful carving through the layers of labradorite brings out flashes of blue, green, yellow, and red as the stone is turned.



Pair of earrings | **Set** | The irregular rounded shapes of these iridescent earrings are set with rows of tiny diamonds around the borders.

Schiller

Lit from within

The iridescence in labradorite is technically called schiller. It is caused by the scattering of light from thin layers of a second type of feldspar that develops through internal chemical separation during the cooling of what was originally a single feldspar. These layers act as diffraction gratings, separating light into its component colours. The colour that results is determined by the thickness of the layers, although the base colour of labradorite is generally blue, dark grey, colourless, or white. High-quality labradorite from Finland is sometimes called spectrolite.



Dramatic coloration
This labradorite specimen has superb schiller.